

● HARD

● CRAFT AND STRUCTURE

Text Structure and Purpose

10 questions · ~15 min

02

Q1

CRAFT AND STRUCTURE

Space scientists Anna-Lisa Paul, Stephen M. Elardo, and Robert Ferl planted seeds of *Arabidopsis thaliana* in samples of lunar regolith—the surface material of the Moon—and, serving as a control group, in terrestrial soil. They found that while all the seeds germinated, the roots of the regolith-grown plants were stunted compared with those in the control group. Moreover, unlike the plants in the control group, the regolith-grown plants exhibited red pigmentation, reduced leaf size, and inhibited growth rates—indicators of stress that were corroborated by postharvest molecular analysis.

Which choice best states the main purpose of the text?

- A. It describes an experiment that addressed an unresolved question about the extent to which lunar regolith resembles terrestrial soils.
- B. It compares two distinct methods of assessing indicators of stress in plants grown in a simulated lunar environment.
- C. It presents evidence in support of the hypothesis that seed germination in lunar habitats is an unattainable goal.
- D. It discusses the findings of a study that evaluated the effects of exposing a plant species to lunar soil conditions.

The following text is adapted from Zora Neale Hurston’s 1921 short story “John Redding Goes to Sea.” John is a child who lives in a town in the woods.

Perhaps ten-year-old John was puzzling to the folk there in the Florida woods for he was an imaginative child and fond of day-dreams. The St. John River flowed a scarce three hundred feet from his back door. On its banks at this point grow numerous palms, luxuriant magnolias and bay trees. On the bosom of the stream float millions of delicately colored hyacinths. [John Redding] loved to wander down to the water’s edge, and, casting in dry twigs, watch them sail away down stream to Jacksonville, the sea, the wide world and [he] wanted to follow them.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It provides an extended description of a location that John likes to visit.
- B. It reveals that some residents of John’s town are confused by his behavior.
- C. It illustrates the uniqueness of John’s imagination compared to the imaginations of other children.
- D. It suggests that John longs to experience a larger life outside the Florida woods.

The following text is adapted from *Indian Boyhood*, a 1902 memoir by Ohiyesa (Charles A. Eastman), a Santee Dakota writer. In the text, Ohiyesa recalls how the women in his tribe harvested maple syrup during his childhood.

Now the women began to test the trees—moving leisurely among them, axe in hand, and striking a single quick blow, to see if the sap would appear. The trees, like people, have their individual characters; some were ready to yield up their life-blood, while others were more reluctant. Now one of the birchen basins was set under each tree, and a hardwood chip driven deep into the cut which the axe had made. From the corners of this chip—at first drop by drop, then more freely—the sap trickled into the little dishes.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It portrays the range of personality traits displayed by the women as they work.
- B. It foregrounds the beneficial relationship between humans and maple trees.
- C. It demonstrates how human behavior can be influenced by the natural environment.
- D. It elaborates on an aspect of the maple trees that the women evaluate.

The following text is from George Marion McClellan's 1895 poem "Eternity."

My spirit swoons, and all my senses cry
For Ocean's breast and covering of the sky.
Rock me to sleep, ye waves, and outward bound,
Just let me drift far out from toil and care,
Where lapping of the waves shall be the sound,
Which mingled with the winds that gently bear
Me on between a peaceful sea and sky,
To make my soothing slumberous lullaby.

Which choice best states the main purpose of the text?

- A. To illustrate the increasing intensity of the speaker's desire to escape ongoing hardship by gliding on the ocean
- B. To contrast the demands of the speaker's everyday life with the serenity of being rocked to sleep by the ocean
- C. To convey the speaker's longing for the ocean to impart a sense of inner tranquility
- D. To justify the speaker's qualms about being transported by the ocean to a quiet destination

Dubautia carrii is a species in a family of plants known collectively as the silversword alliance, all of which grow only on the Hawaiian Islands. Members of this alliance exhibit an extraordinary range of phenotypes, with some species maturing into vines and others into shrubs and trees. All species in the alliance descended from a single ancestral tarweed plant that arrived on the islands around 5 million years ago. The tarweed's descendants diversified into distinct species as they adapted to live in the wide variety of habitats found on the Hawaiian Islands.

Which choice best states the main purpose of the text?

- A. To indicate the specific tarweed ancestor of all plants that grow on the Hawaiian Islands and explain why the plants have such varied physical characteristics
- B. To describe the specific habitat where *Dubautia carrii* are found and identify other plants that share a common ancestor with them
- C. To describe the silversword alliance and explain how the plant family became so varied
- D. To advance the claim that all plants on the Hawaiian Islands are part of the silversword alliance and list possible ancestors of the plants that make up the alliance

In response to concerns that some recent financial crises were exacerbated by consumers misunderstanding risks associated with credit cards, loans, and other financial products, policymakers in many countries have instituted risk-disclosure requirements on sellers of those products. Enrique Seira et al. investigated a variety of risk-disclosure messages sent to thousands of credit card customers and found that the messages had only small and short-lived effects on behavior. Seira et al. asserted that such effects may nevertheless be worth pursuing, given the negligible cost of messaging.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It notes a factor that led Seira et al. to not dismiss risk-disclosure messaging altogether despite their evidence of its limited utility.
- B. It acknowledges a type of risk-disclosure messaging that Seira et al. may not have fully accounted for in their study.
- C. It describes a consideration that explains why Seira et al. recommended risk-disclosure messaging even though its effects may be small relative to its costs.
- D. It points out a circumstance that Seira et al. conceded may make risk-disclosure messaging more effective than their study suggests.

Asteroid 6478 Gault has experienced intermittent mass loss since at least 2013, but in contrast to some other asteroids with repeated mass-loss episodes, 6478 Gault has not lost mass at its perihelion (the closest point of its orbit to the Sun), and thus the loss is not attributable to solar energy–driven ice vaporization. And as Jane X. Luu et al. point out, the singular nature of impact ejection makes it untenable as an account of multiple loss episodes of similar duration over several years. Instead, Luu et al. are likely correct that 6478 Gault is shedding mass due to rotational instability.

Which choice best describes the overall structure of the text?

- A. It presents a scientific observation, describes a contrast between that observation and other observations, and then explains why those other observations should not be considered credible.
- B. It describes an astronomical finding, discusses competing theories about that finding that the author regards as flawed, and then describes new evidence that supports an alternative theory.
- C. It introduces a natural phenomenon, refutes two potential explanations for that phenomenon, and then presents a third explanation for that phenomenon that the author regards as plausible.
- D. It discusses a physical process, evaluates possible causes of that process, and then states that a persuasive account of the process has yet to be put forward.

The mimosa tree evolved in East Asia, where the beetle *Bruchidius terrenus* preys on its seeds. In 1785, mimosa trees were introduced to North America, far from any *B. terrenus*. But evolutionary links between predators and their prey can persist across centuries and continents. Around 2001, *B. terrenus* was introduced in southeastern North America near where botanist Shu-Mei Chang and colleagues had been monitoring mimosa trees. Within a year, 93 percent of the trees had been attacked by the beetles.

Which choice best describes the function of the third sentence in the overall structure of the text?

- A. It states the hypothesis that Chang and colleagues had set out to investigate using mimosa trees and *B. terrenus*.
- B. It presents a generalization that is exemplified by the discussion of the mimosa trees and *B. terrenus*.
- C. It provides context that clarifies why the species mentioned spread to new locations.
- D. It offers an alternative explanation for the findings of Chang and colleagues.

The field of study called affective neuroscience seeks instinctive, physiological causes for feelings such as pleasure or displeasure. Because these sensations are linked to a chemical component (for example, the release of the neurotransmitter dopamine in the brain when one receives or expects a reward), they can be said to have a partly physiological basis. These processes have been described in mammals, but Jingnan Huang and his colleagues have recently observed that some behaviors of honeybees (such as foraging) are also motivated by a dopamine-based signaling process.

What choice best describes the main purpose of the text?

- A. It describes an experimental method of measuring the strength of physiological responses in humans.
- B. It illustrates processes by which certain insects can express how they are feeling.
- C. It summarizes a finding suggesting that some mechanisms in the brains of certain insects resemble mechanisms in mammalian brains.
- D. It presents research showing that certain insects and mammals behave similarly when there is a possibility of a reward for their actions.

Curious about how people visually perceive objects in their dreams, Stephen LaBerge and team recruited lucid dreamers—people aware that they’re dreaming as it’s happening—for a research study. These participants were reliably able to signal when they had entered a dream state; the team then observed participants’ eye movements as they slept. The smoothness with which participants’ eyes tracked objects in their dreams closely matched how sighted people who are awake visually track objects around them, suggesting to the team that the brain perceives dream objects as the product of something other than pure imagination.

Which choice best states the function of the underlined portion in the text as a whole?

- A. To offer key evidence that undermines LaBerge and team’s initial hypothesis
- B. To show the unexpected result that led LaBerge and team to change the focus of their study
- C. To illustrate an important real-world implication of LaBerge and team’s main finding
- D. To identify a comparable circumstance that helps justify LaBerge and team’s conclusion